Artificial Intelligence and Machine Learning Handbook
Author: Dr. Alan Turing Institute

Chapter 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) is defined as the simulation of human intelligence in machines that are progra
The core objective of AI is to develop systems capable of performing tasks that typically require human

The Turing Test, introduced by Alan Turing in 1950, remains a foundational benchmark for assessing w

Chapter 2: Machine Learning Fundamentals

Machine learning (ML) is a subset of artificial intelligence focused on building applications that learn fro

The three primary paradigms of machine learning are:
1. Supervised Learning: Algorithms are trained on labeled datasets where each input is paired with the
2. Unsupervised Learning: Algorithms discover hidden patterns, groupings, or features within unlabeled
3. Reinforcement Learning: An agent learns by interacting with an environment through trial and error, r

Chapter 3: Deep Neural Networks and Transformers

Deep Learning is a specialized subfield of machine learning inspired by the structure and function of the

A Perceptron is the fundamental building unit of an artificial neural network, calculating a weighted sum
Backpropagation is the primary algorithm used to train neural networks, calculating the gradient of the l

The Transformer architecture, introduced by Vaswani et al. in 2017 with the seminal paper "Attention Is

Chapter 4: Ethics, Alignment, and AI Safety

As artificial intelligence systems become increasingly capable and pervasive across critical societal dor

Algorithmic bias occurs when machine learning models inherit, reflect, or amplify systemic prejudices p

AI Alignment refers to the challenge of ensuring that artificial intelligence systems reliably pursue goals